

Introduction

Based on the well-documented benefits of VMMC, the World Health Organization (WHO) has recommended the scale-up of VMMC in high HIV prevalence countries. In 2011, the WHO and UNAIDS set a target of reaching 80% VMMC coverage in 13 priority countries by 2015.

Program planners, policy makers and funders alike are in need of quality cost based data in order to effectively and efficiently implement VMMC programs. The policy-relevance of cost information hinges on it being provided in a timely fashion. But cost studies are slow to implement, which can mean critical evidence is unavailable at the time of decision-making. Additionally, funding limitations restrict the type and quantity of research that can be completed.

One strategy to address these challenges is to use existing research to create econometric models to estimate costs and outcomes of interventions without collecting empirical data. This study is the first in a series of studies which synthesizes a collection of existing studies to create predictive cost models.

Our study utilizes the collective power of existing costing research on VMMC services in Africa to predict the costs of providing VMMC services in a variety of settings in Sub-Saharan Africa.

Methods

Description of data

Our study consists of sixteen VMMC costing studies with empirical data identified through an extensive GHCC literature review including published and unpublished costing studies for HIV interventions.

- Primary data: We requested primary data from several studies meeting our inclusion criteria (facility-level, multi-site studies).
- Secondary data: We extracted secondary data from published studies meeting our inclusion criteria whose authors did not respond or declined to share data.

Data standardization

- Convert databases to standardized format (Stata 15).
- According to author's codebook we identified key variables (cost categories and facility characteristics)
- We standardized input cost categories, based on reference case of GHCC
- We estimated standardized unit cost and concatenate the separate cleaned datasets into one final analytic dataset.

Figure 1. Process to create datasets for the analysis.

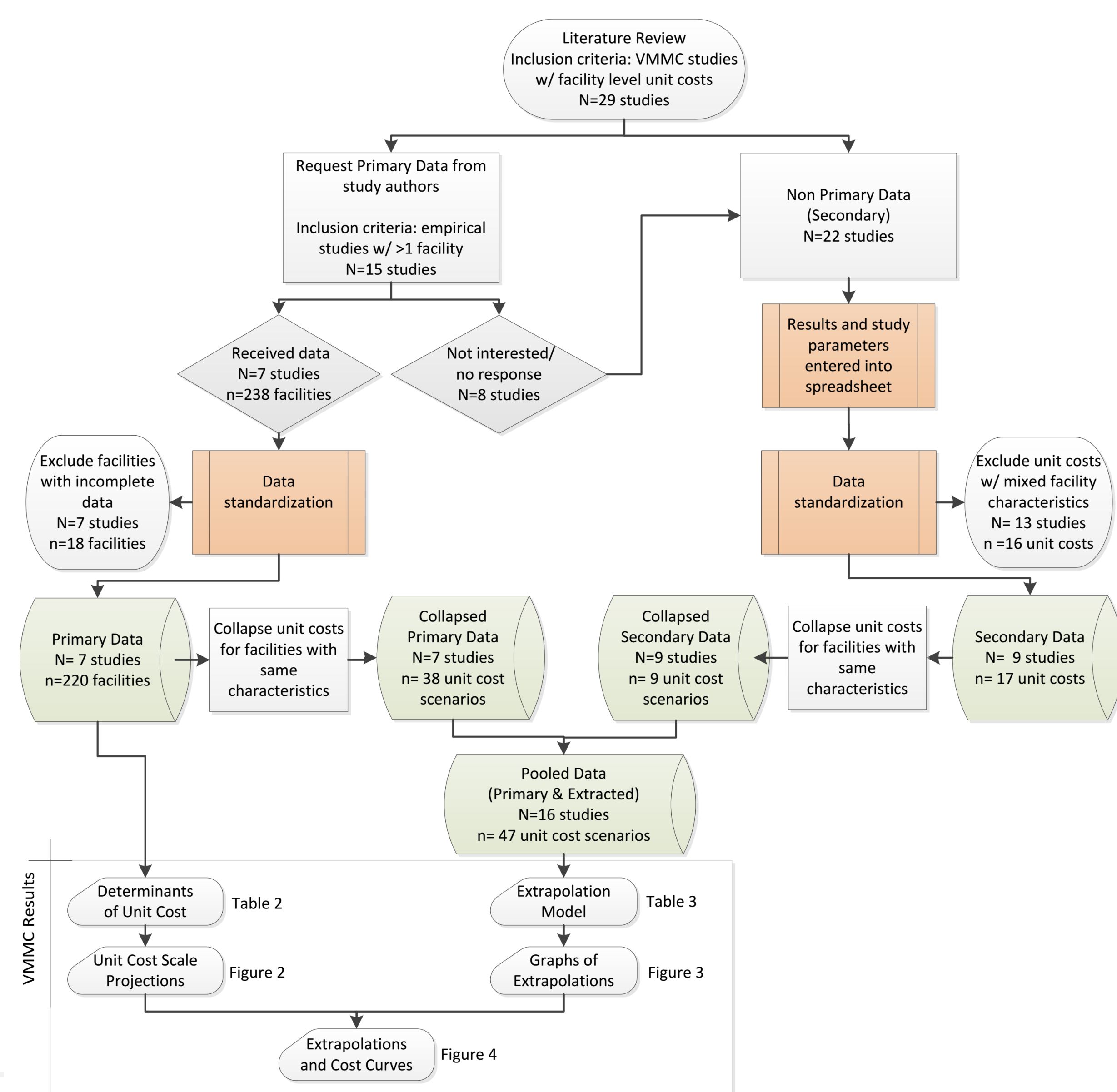


Figure 2. Distribution of facility-level unit costs per VMMC, by country.

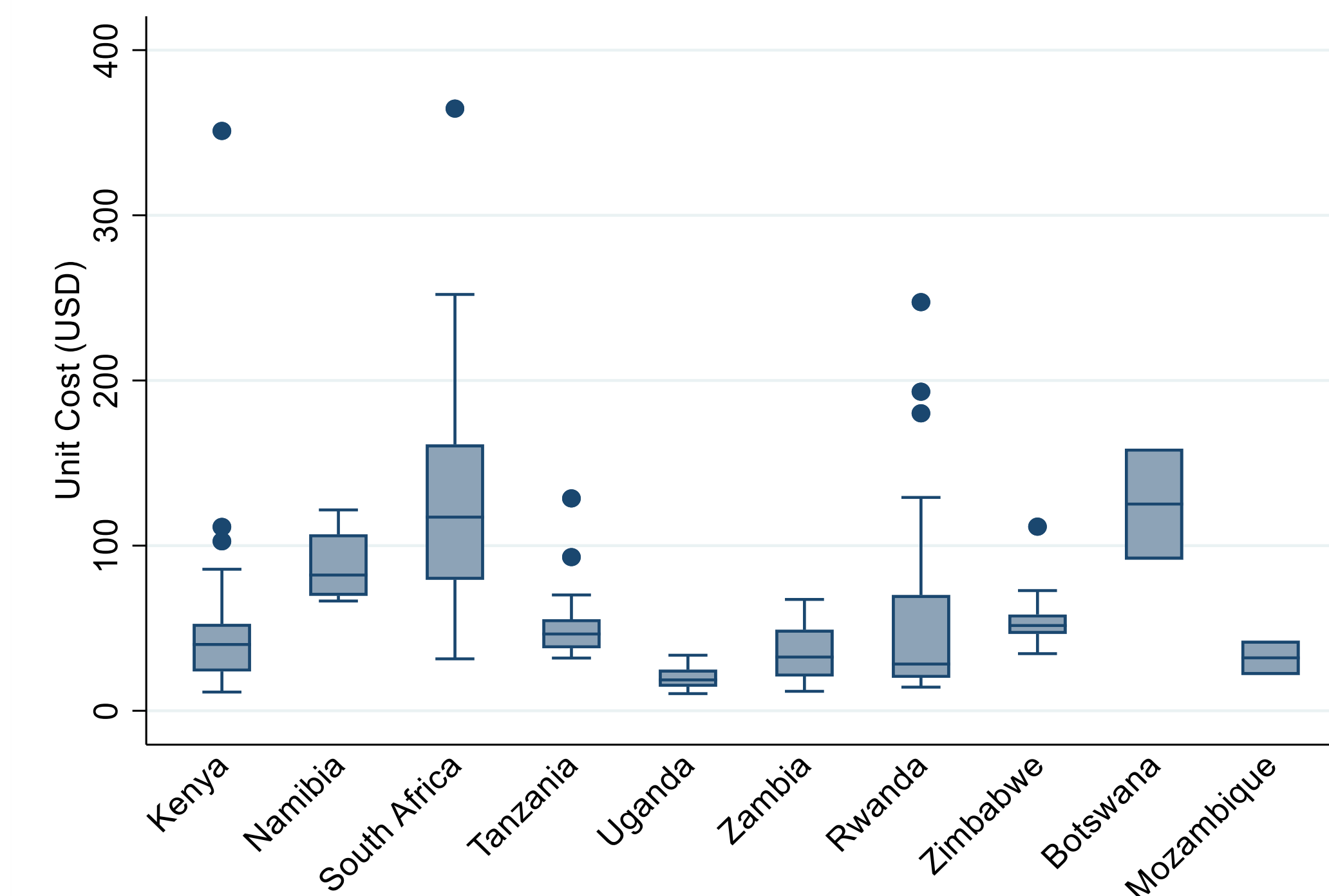


Table 1. Description of Data (primary & secondary)

	Primary Data	Collapsed primary data	Collapsed Secondary data	Pooled data
Observations	220	38	9	47
Urbanicity				
Rural Facilities	50	50	22.2	44.7
Urban Facilities	50	50	77.8	55.3
Ownership				
Private facilities	36.8	36.8	33.3	36.2
Public facilities	63.2	63.2	66.6	63.8
Facility type				
Hospitals	47.3	52.6	66.6	55.3
Clinics	52.7	47.4	33.33	44.7
Mean Unit cost per VMMC in USD (media)	66	64	69	65
Average of VMMC per year	1097	1212	--	--
Average of VMMC coverage (2016)	49	44	30	41
Number of studies	7	7	9	16
Average Number of Facilities per observation	1	6	5.2	5
Countries	8	8	6	10

Analysis with primary data

- The **VMMC unit cost**, defined as the average cost per circumcision, is the total annual cost of male circumcision intervention divided by the total number of male circumcisions performed in the same facility, in the same year
- We conducted OLS and GLM regression models, with fixed and random effects, to estimate the association between facilities characteristics and unit cost variation across sites.
- We used OLS model to predict cost curves with respect to scale.

Analysis with pool data

- We collapsed primary data at the level of implementation scenarios defined by urbanicity, ownership, and facility type, taking the mean unit costs and scale.
- We merged collapsed primary data with secondary data.
- We used OLS models to predict unit cost for countries not included in the sample and validated the predictions by systematically removing data of one country, predicting them, and comparing them with observed values.

Results

Primary data analysis

- We identified facility-level characteristics associated with unit cost variations
- We found results consistent with economies of scale
- Based on coefficients obtained from multivariate regression models (Table 2) we modeled cost curves for each implementation scenario, by country (Figure 3).

Table 2. Determinants of Unit Cost

Dependent variable is VMMC unit cost		
VARIABLES	Fixed	GLM
Scale (VMMC per year)	-0.137*** (0.031)	-0.150*** (0.035)
Scale (VMMC per year) ²	-0.007 (0.014)	-0.008 (0.015)
Rural=1, Urban=0	0.011 (0.079)	0.004 (0.090)
Private=1, Public=0	0.275** (0.125)	0.168 (0.135)
Hospital=1, Clinic=0	0.617*** (0.111)	0.629*** (0.119)
Outreach=1, Fix=0	0.278** (0.121)	0.296** (0.136)
Year of data collection (2016 reference year)	0.146*** (0.029)	0.186*** (0.031)
Hospital*Private	-0.529*** (0.167)	-0.535*** (0.186)
GDP 2016 (log)	0.562*** (0.062)	0.492*** (0.067)
Constant	4.131***	4.495***
Observations	220	220
Standard errors in parentheses		
*** p<0.01, ** p<0.05, *p<0.1		

Figure 3. Unit cost Scenarios

- Each line represents a facility level unit cost scenario defined by urbanicity, ownership, and facility type.
- Dots represent empirically observed facility-level unit costs.

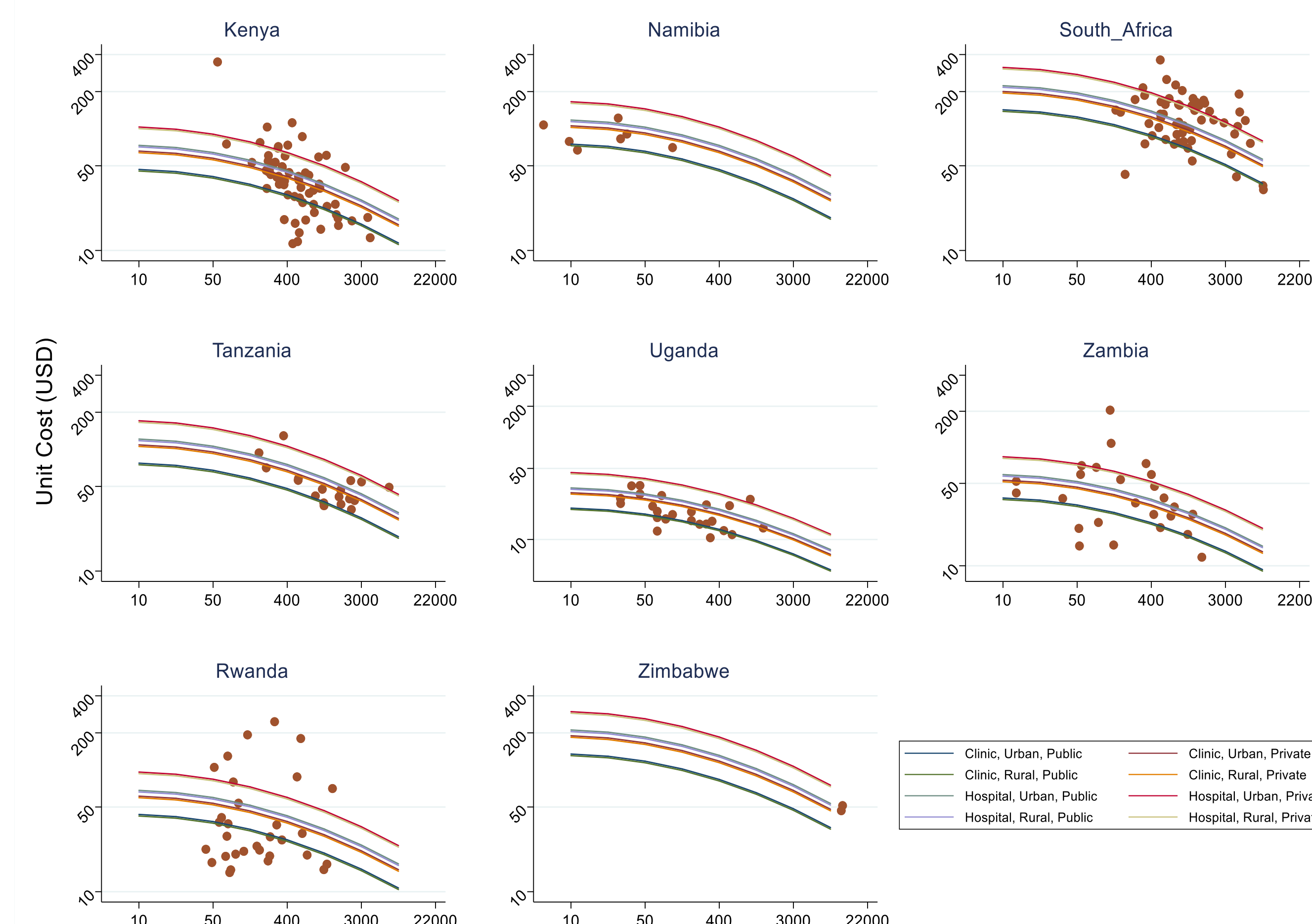


Figure 3. Unit cost Extrapolations

- Comparisons of observed vs predicted unit cost for each scenario by country using pooled data.
- On average we obtained 13% of error in our predictions.

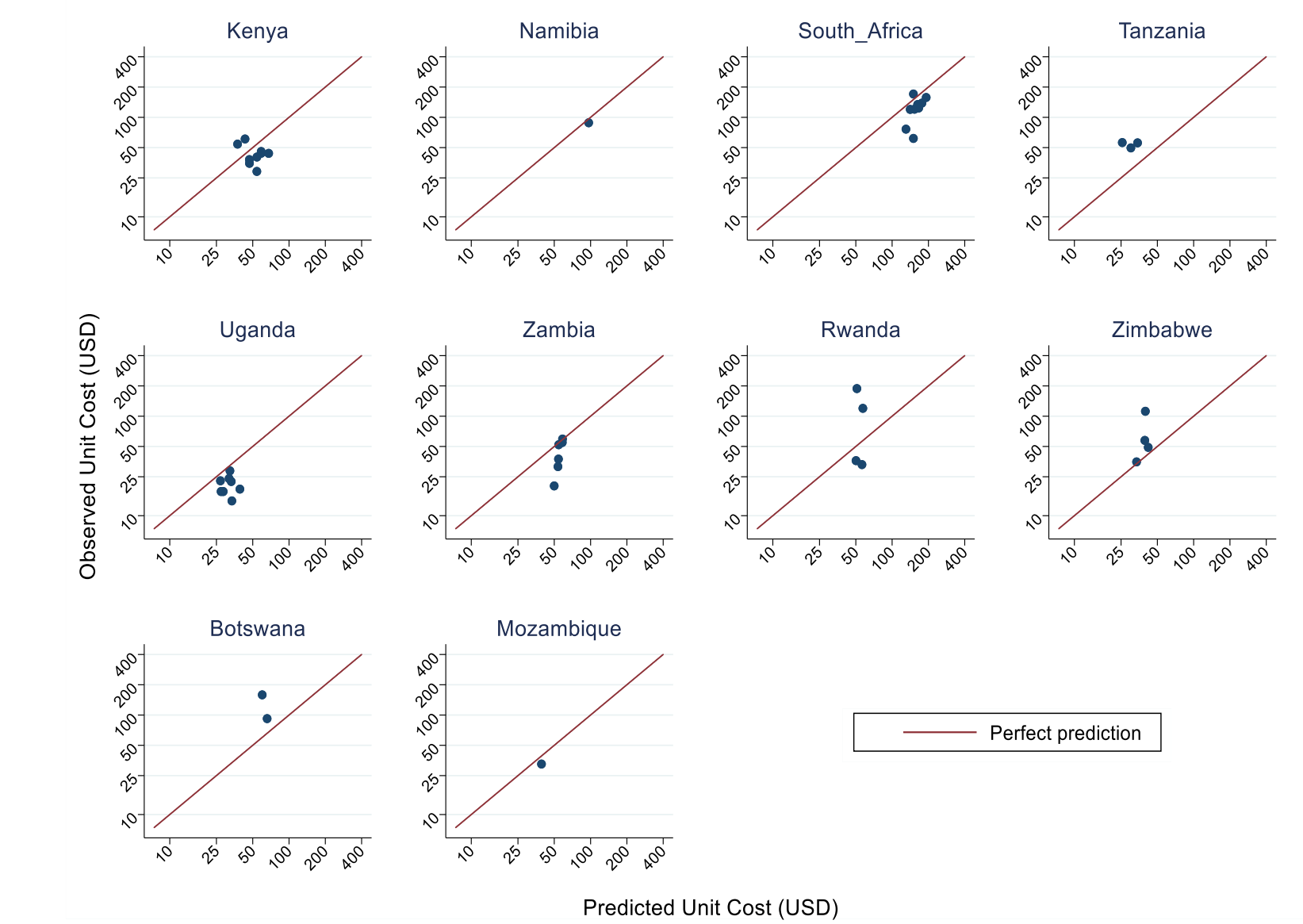
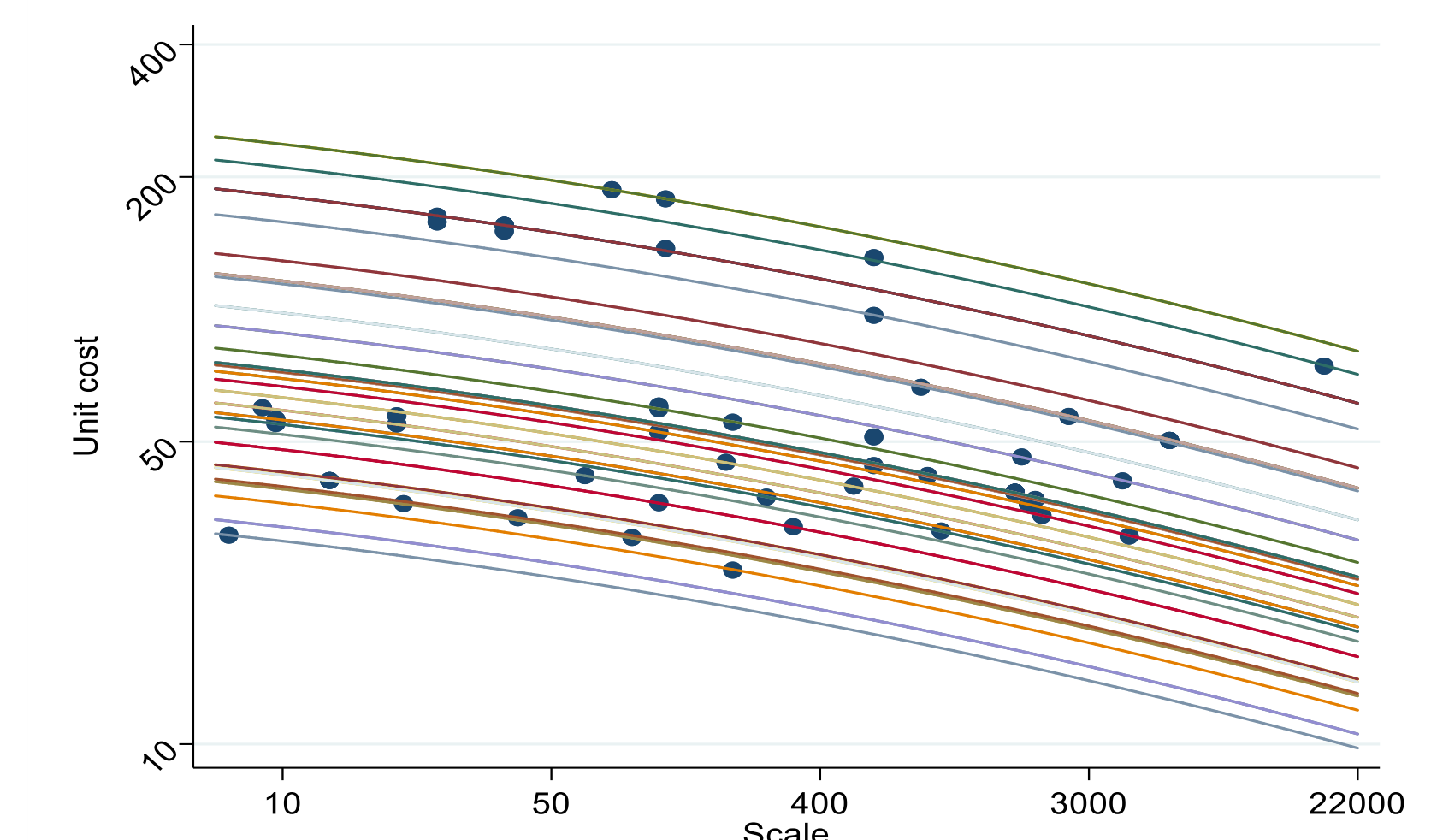


Figure 4. Extrapolated cost curves.

- A combination of data presented in Figures 2 and 3.
- The cost curves represent a predicted unit cost scenario with scale effects.



Conclusions

- We performed VMMC unit cost predictions models, based on primary and secondary data obtained from ten African countries
- We validated our predictions by predicting data out of the sample
- We estimated unit cost curves for different scenarios (Facilities characteristics)
- These models will be part of the inputs of a web tool (UCost) to estimate unit costs of HIV interventions

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